

Web-Based Car Rental System



SE2030 – Software Engineering
2025_Y2_S1_KUR_02

Group members

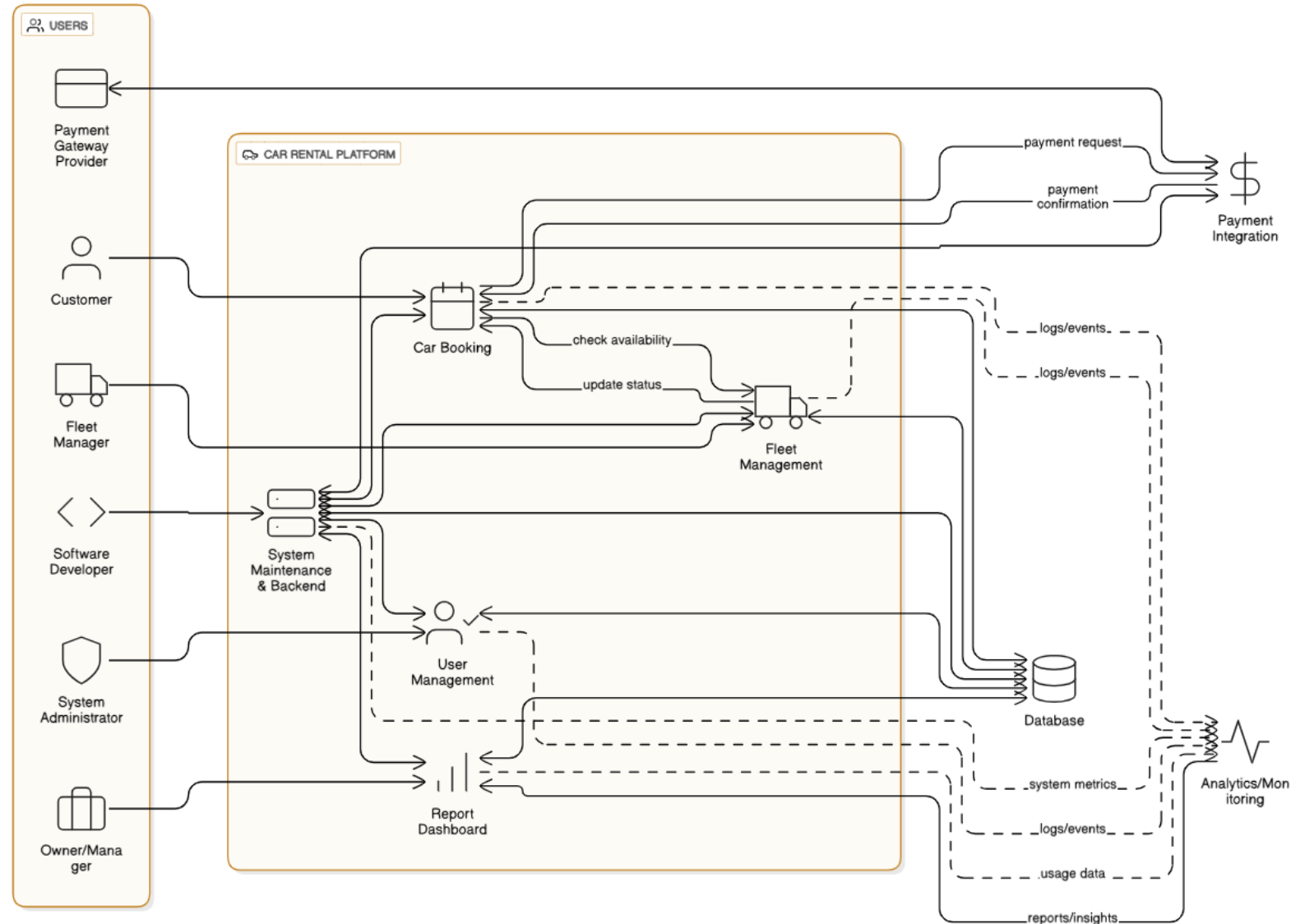
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Introduction

- Developed for SE2030 to replace manual rental tracking with a digital platform.
- **Goals:** Easy car booking, fleet & user management, business insights, and secure payments.
- **Users:** Customers, Admins, Fleet Managers, Owners, Developers, and Payment Providers.
- **Scope:** Booking, user roles, vehicle tracking, reports, payments.
- **Limits:** No GPS, no multi-language (planned for future updates).

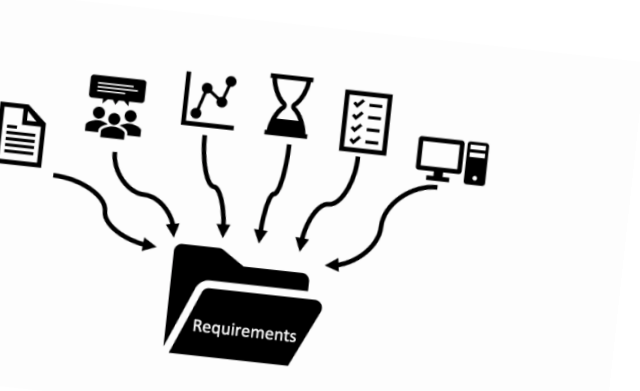


System overview diagram



Functional requirements

1. Book a Car Online Using a Date and Time Selector
2. Manage User Accounts and Permissions (Create, Delete, Update)
3. Create, View and Update Car Availability and Maintenance Status
4. View Business Reports Like Revenue, Bookings, and Usage Trends
5. Maintain Database Connection and Implement Secure Login System
6. Process Online Payments Securely Through Integrated Gateway



Non Functional requirements

1. Usability (e.g.- Clear navigation and feedback)

2. Security (e.g.- Secure payment processing)

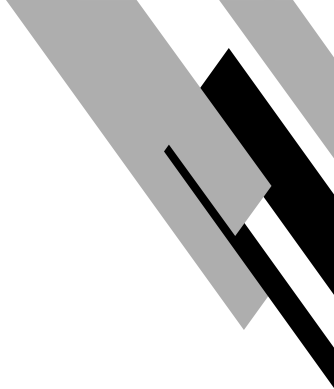
3. Performance (e.g. -Efficient data handling for large datasets)

4. Scalability (e.g. :-Support for multiple users and concurrent bookings)

5. Reliability (e.g. -Persistent data storage to prevent data loss)

6. Maintainability (e.g.- Easy to update or extend functionality)





1.Customer

2.System Administer

3.Fleet Manager

4.Owner Manager

5. Software Developer

6.Payment Gateway Provider



Major stakeholders

Six Major Functions

1. Car Booking System
2. User Management Module
3. Fleet Management System
4. Business Report Dashboard
5. System Maintenance and Backend Support
6. Online Payment Integration



Car Booking System

- Display car details
- Calculate rental costs based on dates
- Get booking confirmation



- Assign roles (staff, customer)
- create, update, delete user accounts
- Log user activity for audits

User Management Module

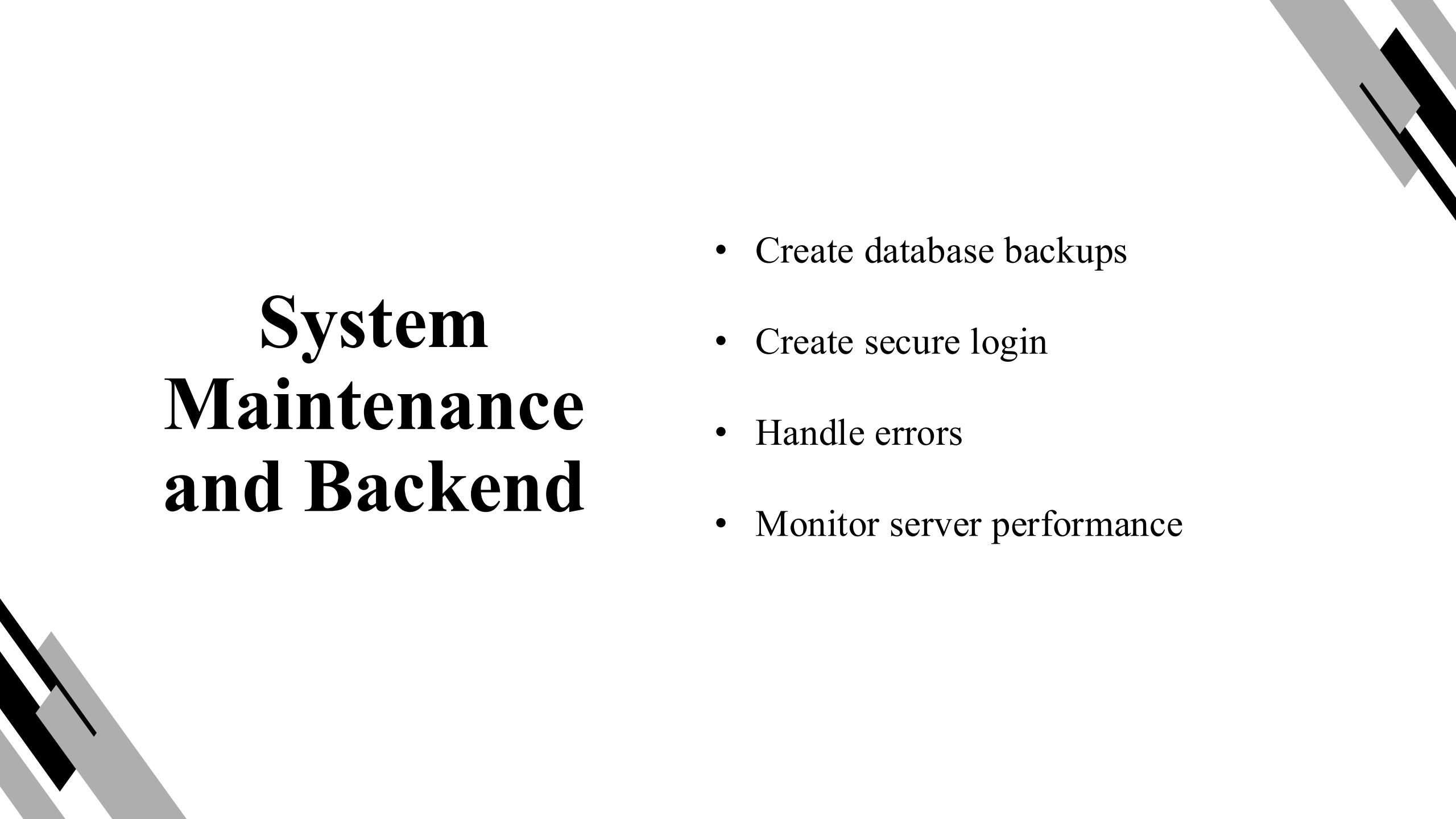
Fleet Management

- create, update and delete vehicles
- Set rental rates for each vehicle

- View customer ratings/comments
- Export reports to PDF/Excel
- Filter reports by time period

Business Report Dashboard





System Maintenance and Backend

- Create database backups
- Create secure login
- Handle errors
- Monitor server performance

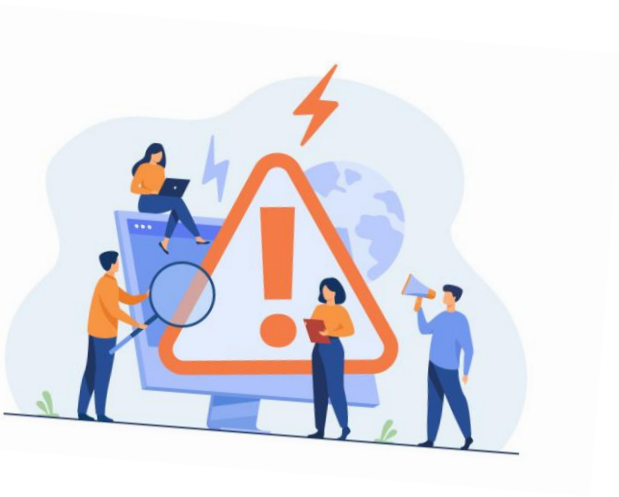
Online Payment Integration

- Store payment history for each booking
- Support multiple payment methods (cash or card)
- Generate receipts



System Limitations

- Platform Dependency
- Device Compatibility:
- Payment Integration Simulation
- Database Limitations
- Internet Dependency
- Internet Dependency



Week	Planned Activities
Week 3	Proposal presentation. Assign roles and core functions to all 6 group members.
Week 4	System design and draft UML diagrams for each member's assigned module.
Week 5	Finalize all UML diagrams
Week 6	Set up backend (Java project, database structure). Assign folder structure for each member's module.
Week 7	- Start to develop Management Module - Start to develop Fleet Management System
Week 8	- Start to develop Car Booking System - Start to develop Online Payment Integration
Week 9	- Start to develop Business Report Dashboard - Start the System Maintenance and Backend Support
Week 10	Progress Presentation.
Week 11	Refine modules based on feedback. Begin integrating modules into one system.
Week 12	Conduct full system testing. Resolve UI/UX issues, handle bugs, and improve performance.
Week 13	Finalize system, prepare documentation
Week 14	Deliver final system and perform Final Presentation

Project Timeline

Conclusion

- We built a modern website for renting cars.
- It's easy for customers, staff, and owners to use.
- Features like live booking make the process faster and more reliable.
- This proves technology is a better way to run the rental business.
- The system is ready to grow and be improved in the future.

Thank You

